1. Aim of the practical: Process and job scheduling commands like ps, stop, kill, fb, bgcrontab, cron etc.

2. Objective: **Automate Task Scheduling**: Schedule the execution of the script.sh located in the /home/aaryaman/ directory at 17:00 (5:00 PM) daily to automate repetitive tasks.

Simplify User-specific Management: Ensure the cron job runs under the correct user (aaryaman) by updating the directory path to reflect the current user's home environment.

Maintain Efficient System Operation: Guarantee timely execution of critical scripts without manual intervention, improving system efficiency and task management.

3. Hardware and Software Requirements:

Computer System: Host system running virtualization software.

Operating System: OS on the virtual machine (e.g., Fedora).

4. Commands/Steps:

Step-1: ps:- Used to check Running Processes

```
asr@fedora:~$ ps
    PID TTY          TIME CMD
   2839 pts/0    00:00:00 bash
   3080 pts/0    00:00:00 ps
asr@fedora:~$
```

Step-2: ps aux used to view Detailed Process Information

```
asr@fedora:~$ ps aux
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.2  1.2  63528 26040 ?        Ss   22:24   0:03 /usr/lib/syst
root           2  0.0  0.0      0     0 ?        S    22:24   0:00 [kthreadd]
root           3  0.0  0.0      0     0 ?        S    22:24   0:00 [pool_workque
root           4  0.0  0.0      0     0 ?        I<   22:24   0:00 [kworker/R-rc
root           5  0.0  0.0      0     0 ?        I<   22:24   0:00 [kworker/R-rc
root           6  0.0  0.0      0     0 ?        I<   22:24   0:00 [kworker/R-sl
root           7  0.0  0.0      0     0 ?        I<   22:24   0:00 [kworker/R-ne
root          13  0.0  0.0      0     0 ?        I<   22:24   0:00 [kworker/R-mm
root          14  0.0  0.0      0     0 ?        I    22:24   0:00 [rcu_tasks_kt
root          15  0.0  0.0      0     0 ?        I    22:24   0:00 [rcu_tasks_ru
root          16  0.0  0.0      0     0 ?        I    22:24   0:00 [rcu_tasks_tr
root          17  0.0  0.0      0     0 ?        S    22:24   0:00 [ksoftirqd/0]
root          18  0.0  0.0      0     0 ?        I    22:24   0:00 [rcu_preempt]
root          19  0.0  0.0      0     0 ?        S    22:24   0:00 [migration/0]
root          20  0.0  0.0      0     0 ?        S    22:24   0:00 [idle_inject/
root          21  0.0  0.0      0     0 ?        S    22:24   0:00 [cpuhp/0]
root          22  0.0  0.0      0     0 ?        S    22:24   0:00 [cpuhp/1]
root          23  0.0  0.0      0     0 ?        S    22:24   0:00 [idle_inject/
```

Step-3: We installed the htop using sudo dnf install htop

```
[sudo] password for asr:
Fedora 40 - x86_64                              2.9 kB/s |  11 kB     00:03
Fedora 40 openh264 (From Cisco) - x86_64        195  B/s | 989  B     00:05
Fedora 40 - x86_64 - Updates                     22 kB/s | 7.8 kB     00:00
Fedora 40 - x86_64 - Updates                    301 kB/s | 6.2 MB     00:20
Dependencies resolved.
================================================================================
 Package            Architecture   Version                Repository       Size
================================================================================
Installing:
 htop               x86_64         3.3.0-3.fc40           fedora          195 k
Installing dependencies:
 hwloc-libs         x86_64         2.11.1-1.fc40          updates         2.1 M

Transaction Summary
================================================================================
Install  2 Packages

Total download size: 2.3 M
Installed size: 3.3 M
Is this ok [y/N]: y
Downloading Packages:
(1/2): htop-3.3.0-3.fc40.x86_64.rpm             129 kB/s | 195 kB     00:01
(2/2): hwloc-libs-2.11.1-1.fc40.x86_64.rpm      257 kB/s | 2.1 MB     00:08
--------------------------------------------------------------------------------
Total                                           219 kB/s | 2.3 MB     00:10
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                        1/1
  Installing       : hwloc-libs-2.11.1-1.fc40.x86_64                        1/2
  Installing       : htop-3.3.0-3.fc40.x86_64                               2/2
  Running scriptlet: htop-3.3.0-3.fc40.x86_64                               2/2

Installed:
  htop-3.3.0-3.fc40.x86_64           hwloc-libs-2.11.1-1.fc40.x86_64

Complete!
```

Step-4: We used (Htop) command to see all the running tasks.

```
  0[|||||                                                          5.6%] Tasks: 126, 382 thr, 126 kthr; 0 running
  1[||||||||||||                                                  13.8%] Load average: 0.21 0.21 0.13
  2[|||||||                                                        7.6%] Uptime: 00:29:44
  3[||||||                                                         6.4%]
Mem[||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||724M/1.92G]
Swp[||||||||||||||||||||||                                     499M/1.92G]

Main  I/O
  PID USER      PRI  NI  VIRT   RES   SHR S  CPU%▽MEM%   TIME+  Command
 2759 asr        20   0 1024M 10132  8980 S   0.0  0.5  0:00.00 /usr/libexec/xdg-desktop-portal
 2764 asr        20   0 1401M  187M 34552 S   0.0  9.5  0:10.98 /usr/bin/gnome-software --gapplication-service
 2765 asr        20   0 1401M  187M 34552 S   0.0  9.5  0:02.87 /usr/bin/gnome-software --gapplication-service
 2769 asr        20   0 1401M  187M 34552 S   0.0  9.5  0:00.00 /usr/bin/gnome-software --gapplication-service
 2785 asr        20   0 1401M  187M 34552 S   0.0  9.5  0:00.00 /usr/bin/gnome-software --gapplication-service
 2796 passim     20   0  312M  8776  8520 S   0.0  0.4  0:00.09 /usr/libexec/passimd
 2797 passim     20   0  312M  8776  8520 S   0.0  0.4  0:00.02 /usr/libexec/passimd
 2798 passim     20   0  312M  8776  8520 S   0.0  0.4  0:00.00 /usr/libexec/passimd
 2799 passim     20   0  312M  8776  8520 S   0.0  0.4  0:00.00 /usr/libexec/passimd
 2812 asr        20   0  769M 12868 11904 S   0.0  0.6  0:00.00 /usr/libexec/ibus-extension-gtk3
 2833 asr        20   0  747M 34020 25908 S   0.0  1.7  0:00.00 /usr/libexec/gnome-terminal-server
 2834 asr        20   0  747M 34020 25908 S   0.0  1.7  0:00.00 /usr/libexec/gnome-terminal-server
 2836 asr        20   0  747M 34020 25908 S   0.0  1.7  0:00.00 /usr/libexec/gnome-terminal-server
 2837 asr        20   0  747M 34020 25908 S   0.0  1.7  0:00.00 /usr/libexec/gnome-terminal-server
 2838 asr        20   0  747M 34020 25908 S   0.0  1.7  0:00.02 /usr/libexec/gnome-terminal-server
 2839 asr        20   0  224M  4992  3456 S   0.0  0.2  0:00.05 bash
 2942 asr        20   0  820M 63212 22272 S   0.0  3.1  0:01.10 /usr/bin/python3 /usr/share/ibus-typing-booster/engine/main.py --ibus
 2943 asr        20   0  820M 63212 22272 S   0.0  3.1  0:00.00 /usr/bin/python3 /usr/share/ibus-typing-booster/engine/main.py --ibus
 2944 asr        20   0  820M 63212 22272 S   0.0  3.1  0:00.00 /usr/bin/python3 /usr/share/ibus-typing-booster/engine/main.py --ibus
 2946 asr        20   0  820M 63212 22272 S   0.0  3.1  0:00.00 /usr/bin/python3 /usr/share/ibus-typing-booster/engine/main.py --ibus
 2947 asr        20   0  820M 63212 22272 S   0.0  3.1  0:00.00 /usr/bin/python3 /usr/share/ibus-typing-booster/engine/main.py --ibus
 2957 asr        20   0  820M 63212 22272 S   0.0  3.1  0:00.00 /usr/bin/python3 /usr/share/ibus-typing-booster/engine/main.py --ibus
 2958 asr        20   0  443M  5760  5376 S   0.0  0.3  0:00.00 /usr/libexec/gvfsd-metadata
 2959 asr        20   0  443M  5760  5376 S   0.0  0.3  0:00.00 /usr/libexec/gvfsd-metadata
 2960 asr        20   0  443M  5760  5376 S   0.0  0.3  0:00.00 /usr/libexec/gvfsd-metadata
 2961 asr        20   0  443M  5760  5376 S   0.0  0.3  0:00.00 /usr/libexec/gvfsd-metadata
 3164 root       20   0  544M 23516 19804 S   0.0  1.2  0:00.05 /usr/libexec/packagekitd
 3165 root       20   0  544M 23516 19804 S   0.0  1.2  0:00.00 /usr/libexec/packagekitd
 3166 root       20   0  544M 23516 19804 S   0.0  1.2  0:00.00 /usr/libexec/packagekitd
 3167 root       20   0  544M 23516 19804 S   0.0  1.2  0:00.00 /usr/libexec/packagekitd
 3179 root       20   0 16524  6144  5376 S   0.0  0.3  0:00.00 systemd-userwork: waiting...
 3180 root       20   0 16524  6144  5376 S   0.0  0.3  0:00.00 systemd-userwork: waiting...
 3189 asr        20   0 4965M  139M 55700 S   0.0  7.1  0:00.01 /usr/bin/gnome-shell
 3190 root       20   0 16524  6144  5376 S   0.0  0.3  0:00.00 systemd-userwork: waiting...
F1Help  F2Setup F3SearchF4FilterF5Tree  F6SortByF7Nice -F8Nice +F9Kill  F10Quit
```

Step-5: Opened a new terminal and killed it using F9 or we can kill using kill <process id>

Step-6: Now installed Crontab (crontab -e)

```
asr@fedora:~$ crontab -e
bash: crontab: command not found...
Install package 'cronie' to provide command 'crontab'? [N/y] y


 * Waiting in queue...
 * Loading list of packages....
The following packages have to be installed:
 cronie-1.7.2-1.fc40.x86_64     Cron daemon for executing programs at set times
 cronie-anacron-1.7.2-1.fc40.x86_64     Utility for running regular jobs
 crontabs-1.11^20190603git9e74f2d-3.fc40.noarch Root crontab files used to schedule the execution of programs
Proceed with changes? [N/y] y


 * Waiting in queue...
 * Waiting for authentication...
 * Waiting in queue...
 * Downloading packages...
 * Requesting data...
 * Testing changes...
 * Installing packages...
```

Step-7: Now Made new executable file for making crontab use.

```
asr@fedora:~$ touch ~/script.sh
asr@fedora:~$ vim ~/script.sh
asr@fedora:~$ chmod +x ~/script.sh
asr@fedora:~$
```

Step-8: Now added this command

```
0 17 * * * /home/asr/script.sh
```

Step-9: Verifying the crontab

```
asr@fedora:~$ crontab -l
0 17 * * * /home/aaryamana/script.sh
```

• Every day at 5:00 PM, the cron daemon (a background service that runs scheduled tasks) will execute the command /home/aaryaman/script.sh. This will run the script you created, which prints "Hello!" to the console.

5. Learning outcomes (What I have learned):

1. Learned to understand process monitoring
2. Learned to understand process control
3. Learned to understand task scheduling

Evaluation Grid (To be created as per the SOP and Assessment guidelines by the faculty):

Sr. No.	Parameters	Marks Obtained	Maximum Marks
1.	Worksheet		8 Marks
2.	Viva		10 Marks
3.	Simulation		12 Marks
	Total		30 Marks

Teacher Signature